

1.1 · Instructor Introduction and Research Talk

Week 2 · Monday Jul 20 — LLMs and Agentic AI
Workflows

Generative AI & Sustainability · Munich Summer
School 2026



CAL POLY



Outline

- Personal and professional introduction
- Overview of my research interests and current work
- Group ice breaker
- Connections: research × generative AI × sustainability × the Week 2 challenge
- Your questions and discussion

Guten Morgen! I'm Alex Bisberg

- **Assistant Professor**, Computer Science & Software Engineering, Cal Poly San Luis Obispo
- Interested in **how people behave in online games and social platforms**
- Studied abroad in **Madrid, Spain** my 2nd year of college
- Traveled to Munich ~10 years ago, only visited for a few days



Academic path: biology → data → games

B.S. Bioengineering

Rice University

Modeled cancer-cell signaling networks — my first taste of large, messy data.

M.S. Computing

University of Utah

Data analytics + first games research: rating systems for esports.

Ph.D. Computer Science

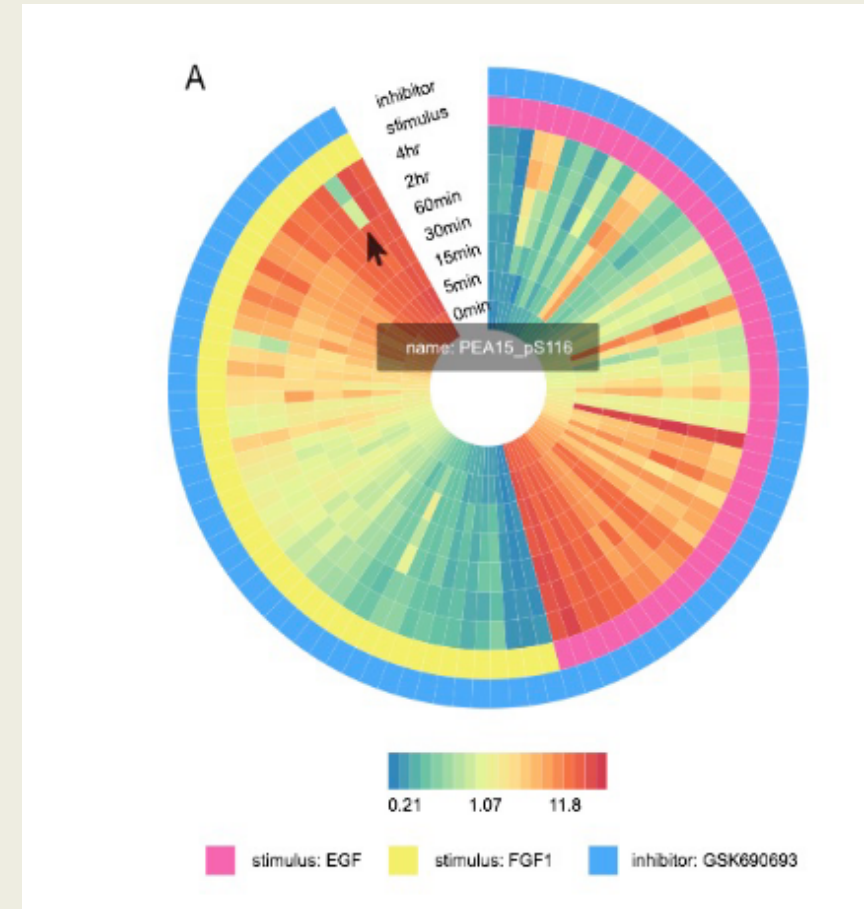
USC

Dissertation: competitive and social success in multiplayer online games.

- "All who wander are not lost"— being multidisciplinary is a strength.

Where it started: proteins, not games

- Undergrad research: **idata visualization** in cancer cells
- Crowdsourced prediction competitions — dozens of teams, one shared dataset
- The skills transferred directly: **feature engineering, prediction, and interactive visualization** (like this explorer we built →)



Industry path: startups, tech, and games



- Founded a start-up, DiBS
- Next: Data Scientist at Harvest Exchange
- During grad school Data science internships at Zillow , Activision Blizzard , Riot Games , Electronic Arts

Games are giant living laboratories

- At Riot: **LoL Esports & Valorant** — do esports lead to more playing?
exposure to toxicity affects play time?
- At EA & Activision Blizzard: bad actor detection
- Telemetry: **who plays, with whom, how they talk, what they give**
 - Every match — like every model — has a **footprint**

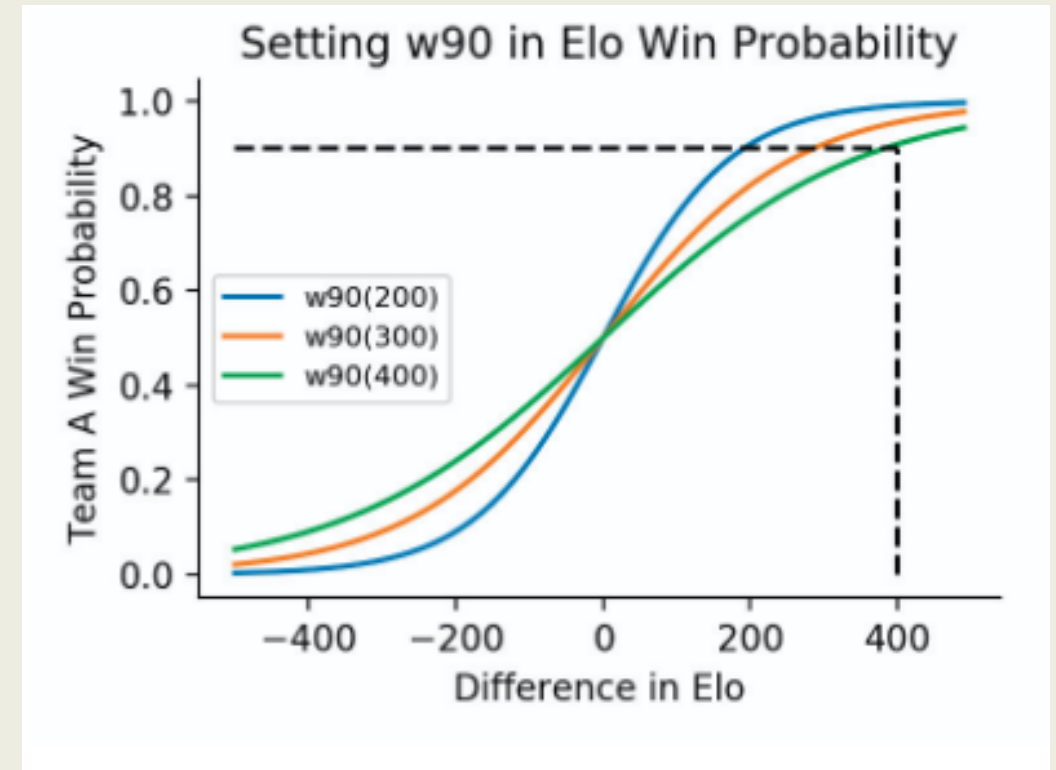


More current research

I use large-scale behavioral data to understand how design choices shape social interaction, inclusion, and long-term engagement in virtual worlds.

Who wins, who talks, who stays?

- **Predicting winners:** Elo rating systems and graph neural networks for esports
- **How teams talk** predicts how well they play
- **Player taxonomies:** sociality, communication, experience, and generosity improve prediction
- Underneath it all: what does **skill** even mean in a team game?



Generosity is contagious

- *Sky: Children of the Light* — a game built around **giving** (candles are the currency of kindness)
- Causal analysis: players who **receive** gifts go on to **give more** (CSCW 2022)
- **Pro-social design** creates self-reinforcing communities



Current work: from social games to social media

- How does perceived **algorithmic personalization** relate to habit formation and problematic use?
- **Osmo**: infrastructure for transparent, reproducible, privacy-respecting social media research
- Long-term goal: **human-first design** — evaluated in the open



Design choices ripple outward

- In games: a gifting mechanic made **generosity contagious**
- In feeds: a ranking tweak changes what millions of people see and feel
- In AI agents: a prompt, a tool, a model choice changes **energy use, cost, and who is served**
- **This week you are the designers. Your agent's choices will ripple too — consider how to measure them.**

Now I want to hear about you!

You've had a week together, I'm the new one here.

Two truths and a hallucination

Round 1 — about me

Which one did the LLM make up?

1. I've jumped out of a plane.
2. I once took down a company's entire website with a change I made.
3. I updated the matchmaking system for League of Legends.

Round 2 — about you

1. Name + home university
2. **One truth, one hallucination** about your Week 1 in Munich
 - I guess which is which...

Where we're headed: Week 2

- **Mon · today** — LLMs & agentic AI workflows
- **Tue** — Data, environment & human-centered AI
- **Wed** — Measurement, evaluation & the pitch
- **Thu** — Final startup pitch to HM

Today's plan

By tonight your team drafts an **agent map** and picks **three agent use cases** for your NextGen learning space.

Questions & discussion

Ask me anything:

- Working in the games industry
- PhD life, research careers, changing fields
- Sky, League, or your favorite game's design
- What I hope you build this week

Contact: abisberg@calpoly.edu



Wrap-up

- Research and games taught me one habit: **measure behavior, then design better systems**
- That habit is exactly what sustainable AI needs — and what your Week 2 challenge rewards
- Up next: **1.2 · From LLMs to Agentic AI Systems** — how these tools actually work
- **Come find me at breaks — I owe each of you a proper hello.**